

BoutScout CSV Format Instructions

Input file requirements for local browser-based incubation inference

1. General rule

BoutScout expects one row per minute. Each complete day must contain exactly 1,440 unique minutes, from 00:00 to 23:59. The app processes the file locally in your browser, groups the data by calendar date, removes duplicated minutes, and runs inference only on complete days.

Incomplete days are excluded automatically. BoutScout does not fill missing minutes, does not interpolate temperatures, and does not repeat previous values. This prevents artificial flat sections at the beginning or end of a day.

2. Required CSV structure

The current web version reads the CSV by column position. The header names below are strongly recommended because they make the file understandable and reproducible, but the app currently uses the order of the columns.

- Column 1 - record_id: recommended. Any row number, logger ID, or record identifier. This column is currently ignored by the app but is useful for traceability.
- Column 2 - datetime: required. Date and time for each minute. This is the most important column.
- Column 3 - egg_temperature: required. Nest, egg, or sensor temperature used as the main signal for inference. Numeric values only.
- Column 4 - ambient_temperature: recommended. Ambient temperature. If this column is missing or not numeric, the app falls back to egg_temperature, which is not ideal for interpretation.

3. Date and time format

Use a clear, unambiguous timestamp. The recommended format is:

YYYY-MM-DD HH:MM:SS

Examples:

```
2024-06-15 00:00:00
2024-06-15 00:01:00
2024-06-15 00:02:00
```

Avoid ambiguous formats such as 06/15/24 or 15/06/24, especially if files will be used across computers with different language or regional settings. Avoid AM/PM unless you have already tested that format locally.

4. Minimal valid example

A valid CSV should look like this. The first row is the header and is skipped by the app during parsing.

```
record_id,datetime,egg_temperature,ambient_temperature
1,2024-06-15 00:00:00,31.82,21.75
2,2024-06-15 00:01:00,31.80,21.76
3,2024-06-15 00:02:00,31.79,21.77
...
1440,2024-06-15 23:59:00,32.10,22.40
```

5. Complete-day rule

BoutScout keeps only days that satisfy all of the following conditions:

- The day has exactly 1,440 unique minute records.
- The first minute of the day is 00:00.
- The last minute of the day is 23:59.
- Each minute appears only once after duplicate removal.
- The date and temperature columns can be parsed correctly.

If a day has 1,439 rows, starts at 00:05, ends at 23:40, contains missing minutes, or has duplicated timestamps, it will be skipped and will not be included in the inference outputs.

6. Temperature values

Temperature columns must contain numeric values only. Do not include degree symbols, commas as decimal separators, units, or text labels inside the temperature cells.

Correct: 31.82
Incorrect: 31,82
Incorrect: 31.82 C
Incorrect: 31.82 deg C

7. Recommended preprocessing before upload

- Sort the file by datetime in ascending order.
- Check that the sampling interval is one minute.
- Remove or review duplicated timestamps before upload.
- Keep only the columns needed for BoutScout, or place the required columns in the first four positions.
- Use a small test file first before uploading many days.
- Do not upload sensitive locality information if the file will be shared publicly.

8. Outputs generated after inference and manual correction

After inference and optional manual editing, BoutScout exports three local CSV files:

- BoutScout_minute_predictions_final.csv: minute-by-minute raw labels and final labels after manual correction.
- BoutScout_events_final.csv: continuous On, Off, Nocturnal, and Unlabeled segments with duration and temperature summaries.
- BoutScout_daily_summary.csv: daily summaries such as nest attentiveness, total On duration, event counts, bout durations, and daily temperature summaries.

9. Practical checklist

- [] CSV file uses comma separators.
- [] First row contains headers.
- [] Column 2 contains datetime.

- [] Column 3 contains nest, egg, or sensor temperature.
- [] Column 4 contains ambient temperature when available.
- [] Each complete day has 1,440 records from 00:00 to 23:59.
- [] No units or text are mixed into temperature columns.
- [] Dates are not ambiguous.

BoutScout runs inference locally in the browser. The input CSV is not uploaded to an external server by the app.